Linear Mixed Models

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Warmup

Work on the warmup activity (handout).

Warmup

Data on 37 undergraduate music majors. Each row represents a performance, and each musician has between 2 and 15 performances in the data.

- `id`: a unique identifier for the musician
- `na`: negative affect score (a measure of anxiety)
- `large`: whether the musician was performing as part of a large ensemble (large $= 1$), or as part of a small ensemble or solo (large $= 0$)

Question: What are the groups in the data, and what are the observations within each group?

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Question: What mixed effects model should the researchers use to investigate their hypothesis?

Models with a random intercept

Intra-class correlation

$$Y_{ij} = u_i + \mathbf{x}_{ij}^T \beta + \varepsilon_{ij}$$

The **intra-class correlation coefficient** (ICC) is the correlation between two observations within the same group i:

$$\rho_{\text{group}} := Cor(Y_{ij}, Y_{ik} | \mathbf{x}_{ij}, \mathbf{x}_{ik}) = \frac{\sigma_u^2}{\sigma_u^2 + \sigma_\varepsilon^2} \quad (j \neq k)$$

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Random effects:
 Groups       Name        Variance Std.Dev.
 neighborhood (Intercept) 1048     32.37
 Residual                 6762     82.23
Number of obs: 1561, groups:  neighborhood, 43
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Parameter estimation

Estimating variance components

Restricted maximum likelihood (REML)

Problem: ML estimates of variance components are biased, because we also have to estimate β

REML solution: Write down a new likelihood function for a transformation of $\mathbf{y}$ that does *not* depend on β

REML